Problem 2.6 of arXiv:1303.4569 was answered negatively by arXiv:1309.0367

Literature-status note

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Status: literature_already_answered. The later paper explicitly identifies the source's Problem 2.6, gives a rank-one real Cartan-factor counterexample, and proves positive results under structural exclusions.

Original question

In *Local triple derivations on C^*-algebras and JB^*-triples* (arXiv:1303.4569), Burgos, Fernández-Polo, and Peralta ask on PDF page 11:

> **Problem 2.6.** Is every bounded local triple derivation on a real JB^*-triple a triple derivation?

Here a local triple derivation is a real-linear map $T : E \to E$ such that for each $a \in E$ there is a triple derivation δ_a with $T(a) = \delta_a(a)$.

Exact later answer

Fernández-Polo, Molino, and Peralta's *Local triple derivations on real C^*-algebras and JB^*-triples* (arXiv:1309.0367) reproduces the question verbatim as Problem 1.1 on PDF page 3, explicitly cites "[8, Problem 2.6]," where [8] is the source paper, and says that it gives a complete answer.

The answer is negative in general. Example 2.4 (PDF page 7) constructs a real-linear local triple derivation on the rank-one type-I Cartan factor $\mathbb{C}^2$, regarded as a real JB^*-triple, which is not a triple derivation. Thus one counterexample settles the universal question.

The same paper also identifies the positive range. Theorem 3.4 (PDF page 10) proves that every local triple derivation on a real JB^*-triple E is a triple derivation when E^{**} contains no rank-one generalized real Cartan factors of types $I^{\mathbb{C}}$, $I^{\mathbb{H}}$, or $V^{\mathbb{O}} = M_{1,2}(\mathbb{O})$. In particular, the paper derives the positive statement for real C^*-algebras.

Provenance and scope

This is an explicit answer, not an agent-inferred theorem match: the supporting authors quote the source problem by number and declare their answer complete. The result resolves Problem 2.6 only. The adjacent module-valued Problem 2.7 in arXiv:1303.4569 asks about continuous local triple derivations into arbitrary Banach triple modules; later work gives special cases, but the searches used for this note did not locate a full solution in that generality.

Searches on 11 August 2026 covered the exact wording of both problems, the source title and arXiv id, local triple derivations on real JB^*-triples, and module-valued variants. The exact cross-citation and complete-answer statement in arXiv:1309.0367 determine the classification.

References

[1] M. Burgos, F. J. Fernández-Polo, and A. M. Peralta, *Local triple derivations on C^*-algebras and JB^*-triples*, arXiv:1303.4569, Problem 2.6 (PDF p. 11).

[2] F. J. Fernández-Polo, A. Molino, and A. M. Peralta, *Local triple derivations on real C^*-algebras and JB^*-triples*, arXiv:1309.0367, Problem 1.1, Example 2.4, and Theorem 3.4.